CrossGuard and GBA AI Quests – Technical Manual

Operator Console, GBA AI Quests and landing page · Version 3.0 · November 2026

Overview

CrossGuard is a proposal for designing, building, auditing and governing a responsible AI system for cross-border deployment in the Greater Bay Area (GBA). It is shown through **two prototypes of one system**, both built on the same crowd model:

- **Run it: the CrossGuard Operator Console.** What a border-crossing operator sees: a camera view of anonymous crowd dots, a 0–100 Crowd Risk Index with its reasons, recommendations a person approves or rejects, AI-labelled notices, a data-boundary panel and an audit log.
- **Learn it: GBA AI Quests.** A 3D map of the Greater Bay Area where every part of the CrossGuard proposal is a quest you can play, plus four hubs and a guided tour that walks through the proposal step by step, in English, Traditional Chinese and Simplified Chinese.

A trilingual landing page links the two. All three are static pages on GitHub Pages:

Page	Address	Source
Landing page	jackyzhijie.github.io/crossguard-gba-ai-quests/	`site/index.html` , `site/main.ts`
Operator Console	…/console/	`site/console/`
GBA AI Quests	…/quests/	`src/`

- **Source code:** https://github.com/JackyZhiJie/crossguard-gba-ai-quests (React 18 + TypeScript, built with Vite)

Every part of the proposal has a place in the prototype where it can be tried:

Proposal part	What it is	Where to try it in the prototype
Design	GBA Responsible AI Harmonisation Model: comply once, certify three times	Three Seals Island quest
Build	CrossGuard: explainable crowd-risk AI for border crossings	Golden Week Surge quest
Audit	Evaluation Matrix and the cross-border interception filter	Open Source Harbour and Filter Forge quests
Govern + educate	TrustLoop: AI literacy, points and gated rewards	Lantau Academy, The Border Gate, Exchange, Hall of Fame

Proposal part	What it is	Where to try it in the prototype
Sustainable loop	Learn → use → expand impact	Exchange, Hall of Fame, Our solution page

Team Bay Builders. Jacky (CityU, MPhil Computer Science): MVP, prototypes and engineering. Billy (PolyU, MSc Building Services Engineering): core idea and crowd-risk model. Winston (HKU, MSc Electrical and Electronic Engineering): system architecture and pilot plan. Thomas (HKU, MSc Energy Engineering): legal harmonisation and governance. Christy (PolyU, MSc Electrical Engineering): problem framing and AI literacy.

Audiences. Reviewers, partners and decision-makers (guided tour and Our solution page), learners from secondary school to early-career professionals (quests and courses), and teachers (Educator guide).

Getting started

Open the landing page in any modern browser on a laptop, tablet or phone and choose **Run it** (the console) or **Learn it** (the game). The rest of this section describes the game. No sign-in or install is needed; it loads React, three.js and fonts from public CDNs.

Screen layout

- **Top bar:** logo (returns to the map), hub buttons (Guided tour, Exchange, Hall of Fame, Educator guide, Our solution), language toggle EN / 繁 / 简, badges earned (out of 6) and your points balance.
- **Quest log (left):** the six quests in suggested order with their status. On a phone it folds into a button at the bottom.
- **Mentor Lumi:** greets first-time players and offers to fly to the first quest.
- **Quest card (right):** appears when a beacon is selected; shows the question the quest answers, the AI concept, time, reward and badge, and a Start button.
- **Quest sheet:** quests and hubs open as a scrolling sheet over the map. "← Map" or Esc closes it.

Controls

Action	Mouse / keyboard	Touch
Rotate the view	Drag	Drag
Zoom	Scroll wheel	Pinch
Pan	Right-drag or Shift + drag	Not available

Action	Mouse / keyboard	Touch
Select a quest	Click a beacon or its label	Tap a beacon or its label
Close a card or sheet	Esc	"← Map" button
Border Gate game	A / R / B / E or 1–4, Enter for next	Tap the buttons

Deep links. Add a hash to the game's address (`…/quests/`) to open a quest or hub directly, for example `#surge`, `#seals`, `#checkpoint`, `#filter`, `#harbour`, `#academy`, `#exchange`, `#hall`, `#educator`, `#about` or `#tour`.

Saved progress. Points, badges, courses passed, shared projects and the chosen language are saved in the browser's local storage. They stay on that device only. Exchange → "Reset demo progress" clears them.

Guided tour. The Guided tour button walks through the proposal in eight stops: each stop flies the camera to the matching beacon and shows the claim, its key points and an "Open this quest" button. Auto-play moves through the stops on its own, so the tour also works as a self-running demo.

Landing page and Operator Console

Landing page (`/`)

One page in the game's visual style with an EN / 繁 / 简 toggle (English by default). It shares the game's language setting (`gba-ai-quests-lang`), so a language chosen in one is used in the other. Sections, in order:

1. **Hero:** "CrossGuard: responsible AI for GBA border crossings", the line "One system, three laws, safer crossings." and two large buttons, **Run it** and **Learn it**.
2. **Two doors:** a card per prototype with a screenshot, who it is for, three things you can do and a button.
3. **How it answers the challenge:** Design, Build, Audit and Govern.
4. **Why now:** three highlights of the 2026 Hong Kong Policy Address, with a link to the official PDF.
5. **What's real and what's simulated:** the five-factor table of the Crowd Risk Index and its assumptions.
6. **Team:** five cards with name, affiliation and role.
7. **Footer:** what the project was built for, the prototype disclaimer and a link to the code.

Text lives in `site/text.ts` (Traditional Chinese written by hand, Simplified converted with OpenCC). Open Graph tags and `img/og.png` give a preview when the link is shared. **Old links keep working:** if the landing page is opened with a game route in its hash (for example `/#surge`, as in earlier slides), `site/main.ts` forwards it to `/quests/#surge`.

Operator Console (`/console/`)

A control-room page in the game's light theme. Its layout, top to bottom:

Area	What it shows
Top bar	Links back to the landing page and to Golden Week Surge; crossing and scenario menus; Play/Pause, Speed ×1/×4, Restart; the simulated clock
Crowd Risk Index	The 0–100 score, its band (Normal under 50, Watch 50–69, Act now 70+), a meter with the alert line, the five factor contributions, and arrivals, throughput and queue
Camera view	An animated top-down view where each dot is one anonymous person (green room to move, amber tight, red packed), e-gate lanes, sensors and a smoke overlay
Risk over the evening	The live index with operator decisions, the “if nobody acts” line, the alert time and the time crowding becomes visible on CCTV
Recommendations	Suggestions once the risk passes 60, each with its reason, to approve or reject; requests outside scope (fire mode) cannot be approved
AI notice	A template-bound trilingual queue notice labelled as AI-generated, which needs operator approval
Data boundary and controls	What stays on each side, what is shared every 15 minutes, and five common controls checked live
Audit log	Every recommendation and decision, copyable as JSON for the joint review panel

Scenarios: Golden Week surge, After Signal 8, Smoke on Platform 2, Sensor spoofing and Friday evening, at Lok Ma Chau, West Kowloon or the HZMB port. **Opening state:** Golden Week at Lok Ma Chau at 17:48, the queue notice already approved and the extra e-gates suggestion waiting.

One model, two prototypes. `site/console/console.ts` imports `step`, `recommend` and `counterfactual` from `src/data/crossguard.ts`, the same module Golden Week Surge uses, so the two prototypes always compute the same Crowd Risk Index. When the console moved from its JavaScript port to this shared module, the two were compared minute by minute for all 3 crossings and 5 scenarios, with and without operator actions: 3,630 simulated minutes, no differences.

Architecture

Everything is static: no server, database or API keys. The game is one HTML page of about 380 KB; React and three.js load from cdnjs and everything else is bundled inline. The landing page and console are plain TypeScript with no framework (about 9 KB and 19 KB of script). Colours, fonts, radii and shadows come from one shared file, `src/theme.css`.

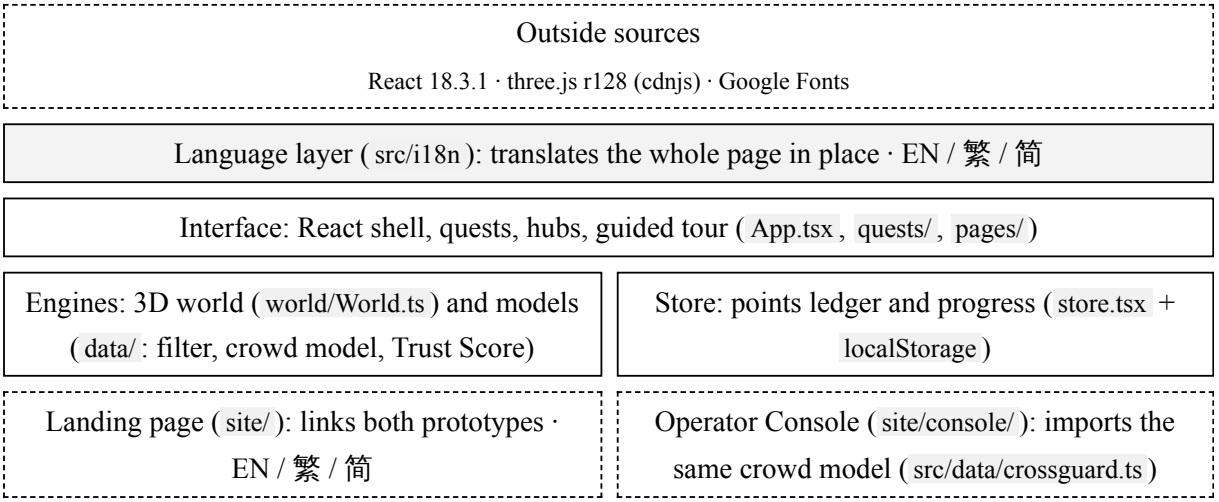


Figure 1. Architecture: the game's three layers inside one HTML page, plus the landing page and console that share its theme and crowd model.

The interface reads the engines and the store; the language layer sits over the whole page and swaps text in place.

Area	Technology	Notes
Language	TypeScript 5.6 (strict)	`tsc --noEmit` runs before every build
UI	React 18.3.1	Classic JSX transform; loaded as the global `React`
3D	three.js r128	Loaded as the global `THREE`; procedural geometry, no model files
Build (game)	Vite 5.4 library mode, IIFE output	`scripts/assemble.mjs` inlines CSS and JS into `dist/quests/index.html`
Build (site)	Vite 5.4 multi-page build (`vite.site.config.ts`)	Landing page and console into `dist/` and `dist/console/`
Fonts	Fredoka, Nunito, JetBrains Mono, Noto Sans TC and SC	Google Fonts with system fallbacks
State	React context + `localStorage`	Key `gba-ai-quests-v1` ; language under `gba-ai-quests-lang`

Source layout

Path	Contents
site/	Landing page (index.html , main.ts , text.ts , landing.css), Operator Console (console/), icons and screenshots (public/)
src/theme.css	Design tokens shared by all three pages
src/App.tsx	Shell: 3D stage, labels, top bar, quest log, quest card, sheets, toasts, badge awarding, hash routing
src/world/World.ts	three.js scene, camera controls, beacons, label projection
src/quests/	Golden Week Surge and Three Seals Island
src/pages/	Border Gate (Play), Academy (Learn), Filter Forge (Guard), Open Source Harbour (Build), Exchange, Ranking, Dashboard, Educator guide, Our solution
src/data/	Content and models: quests, gate cards, courses, projects and matrix, filter engine, CrossGuard model, rewards, dashboard series
src/i18n/	Language layer and generated dictionaries
src/store.tsx	Points ledger, progress, toasts
src/components/	Charts (SVG), mentor, steps, learning tickets, icons, language toggle

Navigation. App.tsx keeps one open value (a quest id, a hub id or none) and mirrors it to the URL hash. Components call go(id) from src/nav.ts , which also maps older names (for example crossguard → surge).

The 3D world

The map is built in code by the World class in src/world/World.ts ; it uses no 3D model files. It is stylised, not to scale: x runs east, z runs south, and one unit is about 2 km.

What is drawn

Element	How it is built
Land	Five extruded polygons (mainland GBA, Hong Kong, Hong Kong Island, Lantau, Macau) with a sand rim
Hills	Low-poly cones scattered with a fixed random seed, kept out of cities and the rail corridor
Cities	One instanced mesh of up to 406 boxes across Shenzhen, Qianhai, Kowloon, Hong Kong Island, Macau and Zhuhai
Water	A 90 × 70 plane whose vertices move with two sine waves each frame
HZMB	A tube along a curve from Lantau to Macau, pylons, three cable towers, two tunnel islands

Element	How it is built
Cross-border rail	Track from Futian to West Kowloon with a four-car train running back and forth
Life	Six boats on circular routes and nine drifting clouds
Beacons	One per quest: base, pulsing ring, light beam, spinning gem, and an invisible hit cylinder for easy clicking. Finished quests turn gold.
Labels	HTML elements (quest names, city names in Chinese and English) positioned each frame by projecting 3D points to the screen

Beacon positions match real places: Lantau Academy (Lantau), The Border Gate (Macau Border Gate), Golden Week Surge (Lok Ma Chau), Filter Forge (Cyberport), Open Source Harbour (Qianhai), Three Seals Island (HZMB artificial island).

Camera. An orbit camera around a target point: azimuth, polar angle (clamped 0.45–1.25 rad) and distance (22–130 units). It turns slowly on its own until the first interaction. Selecting a quest eases the camera to that beacon at distance 42; the guided tour uses the same call. On narrow screens the field of view widens from 42° to 58° so the whole bay fits.

Performance. Pixel ratio is capped at 2, shadows use one 2048 px map, and buildings are a single draw call. Waves, the train, boats, clouds and auto-rotation stop when the system asks for reduced motion.

Fallback. If WebGL is unavailable, the map area shows a short notice and every quest still opens from the quest log.

Quests

Six quests cover the four parts of the proposal: design, build, audit and govern. Each awards a badge once; the beacon and quest log entry turn gold.

#	Quest	Place	Proposal part	Time	Completed when	Badge
1	Lantau Academy	Lantau	Govern (literacy)	10 min	3 of 5 courses passed	Scholar
2	The Border Gate	Macau Border Gate	Govern	5 min	One shift with score ≥ 500	Gatekeeper
3	Golden Week Surge	Lok Ma Chau	Build (CrossGuard)	15 min	Shift ended after passing testing	Crowd Guardian

#	Quest	Place	Proposal part	Time	Completed when	Badge
4	Filter Forge	Cyberport	Audit (filter)	6 min	Sweet-spot setting locked in	Filter Smith
5	Open Source Harbour	Qianhai	Audit (matrix)	6 min	One project shared	Harbour Builder
6	Three Seals Island	HZMB island	Design (harmonisation)	6 min	All three seals stamped	Harmoniser

1. Lantau Academy

Five courses, each three lessons and a three-question quiz; pass with 2 of 3. Courses: AI Literacy 101 (120 points), Data across three systems (150), Prompt safety & API keys (130), Spotting synthetic media (110), Auditing a model (160). Points are paid once per course. Passing Prompt safety unlocks the DeepSeek and Qwen rewards.

2. The Border Gate

A timed card game with 12 cards drawn from a deck of 20: four each from level 1 (data at the border), level 2 (generated content) and level 3 (CrossGuard console). For each card the player chooses Allow, Redact, Block or Escalate within 15 seconds.

- Correct: 100 + up to 50 speed bonus, multiplied ×1.5 after 3 in a row and ×2 after 5.
- A defensible second-best answer: 40.
- Wrong or timed out: lose one of three trust seals; the shift ends at zero.
- Each verdict shows a passport-style stamp, the rule involved (for example PDPO DPP3, PIPL outbound transfer) and the reason.
- Points to wallet = score ÷ 10. Ranks: Trainee, Gatekeeper, Senior Gatekeeper, Chief Border Ethicist (by accuracy).

3. Golden Week Surge

The main quest. The player builds CrossGuard in the six stages of the AI lifecycle, modelled on Google's AI Quests:

1. **Data search.** Pick up to 5 of 9 sources. Five are useful and lawful; social media is noisy; traveller IDs and a face watchlist are privacy breaches; passport nationality is a bias trap.

2. **Data cleaning.** Keep, fix or remove 8 records (a sensor glitch, a missing value, a duplicate, a face image, a UTC timestamp).
3. **Where data lives.** Central server in Shenzhen (fails transfer rules), federated learning (correct), or Hong Kong only (lawful but less accurate).
4. **Model training.** A 30-epoch animation of falling training error.
5. **Model testing.** Predictions are replayed against last Golden Week. Three checks must pass: accuracy ≥ 80%, privacy, fairness. Failing any sends the player back to stage 1, 2 or 3.
6. **Deployment.** Three operator decisions: approve extra e-gates, verify a disagreeing sensor on CCTV, reject a fire-shutter override.

Finish: badge, three certification stamps and three reflection prompts (saved locally). Reward: 400 + 5 per accuracy point above 80 + 30 per correct operator call − 20 per loop back.

4. Filter Forge

The live cross-border filter (formulas in Models and formulas). The player runs sample prompts or types their own, sets origin and destination, and moves the openness–security slider. A chart shows two curves measured on 16 labelled prompts. **Challenge:** find a setting where harmful prompts that pass ≤ 12% and legitimate prompts held ≤ 15% (a narrow window around strictness 50), then lock it in for 200 points. A bypass can be reported for 30 points, up to 3 times. An audit log keeps only a digest of each message.

5. Open Source Harbour

The player enters a GitHub repository, home jurisdiction, risk tier and maturity, and ticks the controls the project has. A simulated six-step audit runs, then the project joins the Evaluation Matrix with its Trust Score, radar chart, tier gaps, three certification stamps and an openness-vs-security release slider. Reward: 150 + 3 per Trust Score point above 50; later “Sync stars” pays 2 per new star, and “Pitch to an organisation” (Hall of Fame) pays 500 once.

6. Three Seals Island

1. **Classify** four AI uses into Minimal, Limited, High or Unacceptable risk.
2. **Map** four obligations (data transfer, transparency, human oversight, generated content): for each, pick the strictest of the Hong Kong, Macau and mainland versions. The common control appears under each.

3. **Certify**: stamp the Hong Kong, Macau and mainland seals.

Reward: 150 + 15 per correct answer (up to 270).

Hubs

Five hubs sit in the top bar and open as sheets over the map (the Guided tour runs on the map itself).

Hub	Purpose	Main functions
Guided tour	Walk visitors through the proposal	8 stops, one per claim; camera flies to each beacon; claim, key points, "Open this quest"; Previous / Next; Auto-play
Exchange	Spend points; show the economy	Wallet balance and total earned, ways to earn, 7 gated rewards with inline confirmation, demo voucher codes with copy, ledger of the last 14 entries, reward-pool funding split, reset progress
Hall of Fame	Show who is learning and what is adopted	Leaderboards (learners with podium and your rank, open-source projects sortable by impact, stars, downloads or trust, adopting organisations with hours saved) and the Impact dashboard
Educator guide	Run AI Quests in a 55-minute lesson	Subjects, audience, duration, enduring understanding, 5 learning objectives, 8 vocabulary terms, lesson flow with timings, the six-task strip, teacher tips, probing questions, reflection prompts, exit ticket, alternative set-ups, extensions
Our solution	The proposal in one page	Thesis, the four parts with links to their quests, the 150-word proposal, the sustainable loop, harmonisation steps, common controls table, rollout, guided tour button

Impact dashboard. Region filters (Hong Kong, Mainland GBA, Macau) and a 4 / 8 / 12-week range drive five KPIs, a weekly active learners line chart with hover tooltip and table view, the learn-to-adoption funnel, points minted vs redeemed, the Trust Score distribution, cumulative GitHub stars and your live contribution. Pilot figures are a simulated 12-week projection and are labelled as such.

Projects in the matrix. Twelve illustrative open-source projects (for example CrossGuard-Edge, pdpo-lint, CantoGuard, synth-label) plus any the player shares.

Points, badges and gated rewards

Every player starts with a 100-point welcome bonus. All earning and spending is written to one ledger in the store, so the balance is the sum of ledger entries.

Earning

Source	Points
Border Gate shift	score ÷ 10
Course passed (first time)	110–160
Project shared	150 + 3 per Trust Score point above 50
New GitHub stars (Sync stars)	2 per star
Organisation adopts your project	500, once per project
Golden Week Surge	400 + bonus
Three Seals Island	150 + 15 per correct answer
Filter Forge sweet spot	200
Filter bypass reported	30, up to 3
Badge for quests 1, 2, 4, 5	100 each

Rewards. Access grows with skill: every reward has a gate, and the button stays locked until the gate is met and the balance covers the cost.

Reward	Cost	Gate	Guardrail
Qwen API credits (500k tokens)	450	Prompt safety course	Spend cap, 30-day expiry, usage logged
DeepSeek API credits (1M tokens)	600	Prompt safety course	Spend cap, 30-day expiry, usage logged
GBA Responsible AI Practitioner badge	800	All 5 courses	Revocable, lists modules passed
GPU compute (10 hours)	1,200	A shared project with Trust Score 60+	Tied to a public repo; jobs logged
Gemini Pro access (1 month)	1,500	AI Literacy 101 and Prompt safety	Personal, non-transferable
Incubation fast-track	3,000	Project at Pilot maturity with Trust Score 70+	Panel reviews the audit report
HK$50,000 seed grant pitch	5,000	Same as incubation	Milestone-based, audited

Redeeming shows a `DEMO-XXXX-XXXX` code generated from a hash. Codes do not work with any provider, and brand names show the kind of reward proposed, not confirmed partners.

Badges. Scholar, Gatekeeper, Crowd Guardian, Filter Smith, Harbour Builder and Harmoniser. `App.tsx` derives completion from the store and awards each badge once, so replaying a quest never pays twice.

Models and formulas

All models are deterministic TypeScript functions in `src/data/`, so the same inputs always give the same result and anyone can reproduce every number.

Cross-border filter (`guard.ts`)

The filter scans text with 15 regular expressions, then decides from the highest severity found and the strictness slider s (0 = open, 100 = secure), where k = s ÷ 100.

Category	Patterns (severity)
Personal data	HKID, mainland ID, Macau ID (0.9); Macau, Hong Kong and mainland phone numbers (0.6); email (0.5)
Prompt injection	Override instructions (0.95), role hijack (0.85), secret extraction (0.55)
Unverified safety claim	Border closure claim (0.6), typhoon or rainstorm signal claim (0.5)
Deepfake request	Impersonate a real person (0.8)
Violence / self-harm	0.9
Discrimination	Group-based exclusion (0.7)

$$\text{block}_k = 0.98 - 0.48k \qquad \text{flag}_k = 0.90 - 0.60k \qquad \text{redact}_k = 0.45 - 0.35k$$

$$\text{risk}_{\text{PII}} = \min\left(1,\ w \sum \text{severity}_{\text{PII}}\right), \qquad w = 0.7 \text{ cross-border},\ 0.5 \text{ same jurisdiction}$$

1. If the highest non-personal severity ≥ block, the message is blocked.
2. Else if it ≥ flag, it is escalated to a human.
3. Else, personal data: cross-border with PII risk ≥ block + 0.3 is escalated to the DPO; PII risk ≥ redact is masked (each identifier replaced by its label, for example `[HKID]`) and passed.
4. Otherwise it passes and is logged.

The trade-off curve runs the same function over 16 labelled prompts (9 harmful, 7 legitimate) at 21 strictness steps. Sample results: at 20, 33% of harmful prompts pass and 0% of legitimate ones are held; at 50, 11% and 0%; at 70, 0% and 43%.

CrossGuard crowd model (`crossguard.ts`)

A one-minute step model per crossing (Lok Ma Chau 140 arrivals/min and 24 gates; West Kowloon 110 and 20; HZMB 90 and 16). Scenarios change arrivals (Golden Week ramps to 1.8×, Signal 8 lifting adds a decaying backlog) or capacity (smoke halves exit throughput).

$$\text{queue}_t = \max(0,\ \text{queue}_{t-1} + \text{arrivals}_t - \text{throughput}_t), \qquad \text{density}_t = \min\Big(7,\ \frac{\text{queue}_t}{\text{area}} + 0.6\Big)$$

$$\text{CRI} = \min\left(100,\ 6 + 36\,c\big(\tfrac{d-0.6}{3.4}\big) + 24\,c\big(\tfrac{r-0.95}{0.7}\big) + 20\,c\big(\tfrac{d+10\Delta d-1.2}{2.8}\big) + 14\,\text{exits} + 16\,\text{smoke}\right)$$

Here c clamps to 0–1, d is sensed density, r is arrivals over normal and Δd is the density change per minute. The five terms are the factor contributions shown to the operator. The alert threshold is 70; manual monitoring is taken to notice a crowd at 3.5 persons/m². In surge scenarios the index crosses 70 between 10 and 18 minutes before density reaches 3.5. Golden Week Surge replays the Lok Ma Chau Golden Week run for its testing stage.

Trust Score and certification (`projects.ts`)

Seven dimensions are scored 0–5 with weights: transparency 1.2, data localisation 1.3, human oversight 1.3, content safety 1.0, robustness 1.1, fairness 1.0, openness 0.8.

$$\text{Trust} = \text{round}\left(100 \cdot \frac{\sum_i w_i s_i}{5 \sum_i w_i}\right)$$

Check	Rule
Limited tier minimum	Transparency 3, content safety 3
High tier minimum	Transparency 4, localisation 4, oversight 4, robustness 3, fairness 3
Hong Kong certifies	Average of oversight, transparency, fairness ≥ 3.5, each ≥ 3
Macau certifies	Average of localisation, transparency, robustness ≥ 3.5, each ≥ 3
Mainland certifies	Average of localisation, content safety, transparency, robustness ≥ 3.75, each ≥ 3
Release ceiling	Minimal: fully open · Limited: weights open · High: code open · Unacceptable: not listed

A shared repository is scored as 0.8 + a 0–0.8 value from a hash of its URL, plus points for each ticked control (for example a model card +2 transparency, local processing +3 localisation), +0.5 robustness and transparency at maturity L3 or above, capped at 5 and rounded to 0.5.

Golden Week Surge accuracy (`Surge.tsx`)

$$\text{accuracy} = \text{round}\left(\max\left(40,\ 50 + 40\,\min(1, U)\,(0.55 + 0.45\,C)\,A - N\right)\right)$$

U is the useful signal chosen over the maximum of 12, C is the share of records cleaned correctly, A is 1 (or 0.82 for Hong Kong only), and N is 3 if social media was chosen. The best possible run scores 90%.

Languages

The whole interface switches between English, Traditional Chinese (Hong Kong wording) and Simplified Chinese from the EN / 繁 / 简 toggle in the top bar and in every sheet. The choice is saved per device.

How it works. Components are written in English. The language layer in `src/i18n/index.ts` swaps visible text in place:

1. A tree walk translates every text node and the `placeholder`, `title` and `aria-label` attributes, remembering each node's English source.
2. A `MutationObserver` translates anything React adds or changes afterwards, so live values (timers, scores, quest steps) stay translated.
3. Exact matches come from the dictionary; about 30 patterns handle text built from numbers or names (for example "Quest complete: …", "Risk 82: …", "3 of 5 courses passed").
4. Switching back to English restores the remembered source text.
5. In Chinese modes, CSS hides the small bilingual labels so names do not appear twice, and switches the CJK font to Noto Sans TC or SC.

Dictionaries. `src/i18n/strings.ts` holds about 1,180 English → Traditional entries written by hand, the Simplified set generated from them with OpenCC (Hong Kong → mainland conventions), and a character map for Chinese already in the source. Project and brand names (CrossGuard, TrustLoop, DeepSeek, GitHub) and typed sample prompts stay in their original language.

Adding or changing a translation. Edit the tables in `i18n-tools/t1.py` to `t5.py`, run `bash i18n-tools/regen.sh` to regenerate the dictionaries, and rebuild. New English text that is not in the dictionary simply shows in English until it is added.

Privacy, data handling and accessibility

The prototype practises what the proposal asks of CrossGuard: no personal data leaves the device.

- **No accounts, no server.** Nothing typed into the prototype is sent anywhere. The only network requests are for React, three.js and fonts.
- **Local storage only.** Progress, reflections and the language choice are kept in `localStorage`; every read and write is wrapped so the page still works when storage is blocked (private windows).

- **Filter audit log.** Keeps time, route, decision, rules and an 8-character digest, never the message text.
- **Sample personal data is fictional.** Names, ID numbers and phone numbers in prompts and cards are made up for the exercises.
- **Simulated audits.** The GitHub audit never contacts GitHub; it derives a repeatable score from the URL and the ticked controls.

Accessibility

- Every quest is reachable from the quest log without using the 3D map, and the map has a WebGL fallback.
- Keyboard: Esc closes, the Border Gate game has letter and number shortcuts, and focus outlines are visible.
- Charts have text labels, legends, hover values and a table view on the dashboard; status is never shown by colour alone.
- Animations stop under the reduced-motion setting.
- Layouts reflow to phone width; the quest log and quest cards move to the bottom of the screen.

Build, deploy and test

Requirements: Node 18 or later and npm. Python 3 is needed only for regenerating translations.

Command	What it does
`npm install`	Installs React, three.js typings, TypeScript and Vite
`npm run dev`	Starts the game with hot reload
`npm run dev:site`	Starts the landing page and console with hot reload
`npm run typecheck`	Runs `tsc --noEmit` (strict mode)
`npm run build`	Cleans `dist/`, type-checks, builds the game into `dist/quests/`, then builds the landing page and console into `dist/`
`npm run deploy`	Builds, then publishes `dist/` to the `gh-pages` branch with the `gh-pages` package
`bash i18n-tools/regen.sh`	Regenerates `src/i18n/strings.ts` from the translation tables (run `npm install --no-save opencc-js@1.0.5` once first)

Output. `dist/index.html` (landing page), `dist/console/index.html` (console) and `dist/quests/index.html` (game, about 380 KB, with its icons and app manifest). React 18.3.1, ReactDOM 18.3.1 and three.js

r128 are not bundled; the game loads them from cdnjs as globals, and fonts come from Google Fonts. All links are relative, so the site works under the `/crossguard-gba-ai-quests/` base path. **Deployment:** every push to `main` runs the GitHub Actions workflow `.github/workflows/deploy.yml`, which installs dependencies, builds the TypeScript source and publishes `dist/` to the `gh-pages` branch served by GitHub Pages; `npm run deploy` does the same from a local machine.

Deep links. Every quest and hub has a hash URL under `/quests/`, so documents and presentations can link straight to it: `#surge`, `#seals`, `#checkpoint`, `#filter`, `#harbour`, `#academy`, `#exchange`, `#hall`, `#educator`, `#about` and `#tour`.

Testing done. Automated browser runs with Playwright (headless Chromium, software WebGL) checked that:

- the map renders and every beacon opens its quest card, with no console errors;
- each quest can be completed end to end and pays its points and badge;
- the language toggle round-trips EN → 繁 → 简 → EN, including in the middle of a quest, and leaves only project names in English;
- the guided tour steps through all eight stops, opens each linked quest and returns to the tour;
- the WebGL-off fallback still lists every quest;
- layouts hold at phone width (390 px);
- at 1280 px and 390 px, `/`, `/console/`, `/quests/` and `/quests/#surge` load under the `/crossguard-gba-ai-quests/` base path with no console errors and no horizontal scrolling, and `/#seals` forwards to `/quests/#seals`;
- the landing page language toggle switches all text, and every link on it returns a page;
- the console opens in its documented state, and approve, reject, the blocked fire-mode request, the spoofed-sensor check and the audit-log copy all work.

There is no unit-test suite yet; the models are small pure functions in `src/data/*`, so they are the easiest place to add one.

Limitations, simulated content and how to extend

What is simulated. The prototype demonstrates CrossGuard and its governance loop; it is not connected to real systems.

- Both prototypes run on simulated crowd data. The Crowd Risk Index is a transparent weighted score of five factors; density counts fully at 4 people per m² (the UK Green Guide limit for holding areas), and the other weights are assumptions to be fitted to real data in a pilot.
- Crowd data, risk forecasts, pilot figures and the 12-week dashboard projection are generated by the models in `src/data/`, not measured at a real crossing. The "10+ minutes earlier" figure is the pilot hypothesis.
- GitHub audits, stars and downloads in Open Source Harbour are simulated from what the player enters; no repository is fetched.
- Projects, people and organisations in the Hall of Fame are illustrative. Partner brands in the Exchange show the kind of reward proposed, not confirmed partnerships, and vouchers are demo codes.
- The legal summaries for Hong Kong, Macau and the mainland are simplified for teaching and are not legal advice.
- Progress lives in one browser. There is no backend, so rankings compare the player with sample entries only.

How to extend.

- **Add a quest:** add an entry to `src/data/quests.ts` (id, title, place, position, colour, reward), write its component in `src/quests/`, register it in `App.tsx`, and add a badge rule if it should award one. The beacon appears on the map automatically.
- **Add a translation:** add the English → Traditional pair to `i18n-tools/t5.py` (`EXTRA`), run `bash i18n-tools/regen.sh`, rebuild. For text built from numbers, add a pattern in `src/i18n/index.ts`.
- **Add or change a tour stop:** edit `TOUR` in `src/data/tour.ts`. Each stop has a title, key points, a camera target (`focus`), an optional quest or hub to open and its auto-play time in `seconds`.
- **Change the proposal summary:** edit `PROPOSAL` in `src/data/tour.ts`; the Our solution page reads it.

- **Change the crowd model:** edit `src/data/crossguard.ts`; both the game and the console pick up the change on the next build.
- **Change the landing page text:** edit `site/text.ts` (all three languages) and the matching `data-i18n` keys in `site/index.html`.
- **Go beyond a demo:** a production version would add a backend for accounts and the points ledger, real GitHub OAuth for project audits, and feed the operator console from on-site edge counters.